

DSA4266 Group 2 Proposal

Problem Statement

This project investigates whether it is feasible to build a low-latency, low-parameter Domain Generation Algorithm (DGA) detection system that maintains strong performance without failing when encountering new or unseen malware families.

In other words: Are we building durable detection systems, or merely optimizing for static datasets that fail when new malware appears?

Our dataset, sourced from Kaggle (Givre, 2021), consists of 160K labeled domain strings (80K benign, 80K DGA), with DGA samples categorized into malware subclasses. Since only raw domain strings are available, detection must rely purely on lexical patterns. While minimizing False Positive Rate (FPR) and inference latency is important for deployment, the broader concern is model robustness under evolving adversarial behavior.

Approach

We adopt a progressive modeling strategy that prioritizes interpretability of trade-offs over raw benchmark performance. Rather than optimizing for a single metric, the approach is designed to surface the tension between efficiency and generalization — asking at each stage whether added complexity genuinely improves robustness or merely inflates performance on seen data. Informed by prior work on DNS traffic analysis and ML-based DGA detection (Manasrah et al., 2022; Jeremiah et al., 2025), we begin with a simple, efficient classifier to establish a performance baseline, then selectively increase model capacity only when justified by measurable gains in generalization. Each modeling decision is evaluated against its operational cost in latency and parameter count, ensuring the final system is deployment-viable rather than benchmark-tuned.

Methodology & Exploration

With previous works in mind, we begin with a simple and efficient classifier to establish reference performance in False Positive Rate, detection accuracy, inference latency, and model size. This creates a reference point against which all subsequent complexity is justified. We then gradually increase model capacity only when necessary to improve generalization across malware families, stability under distribution shifts, and resistance to overfitting known DGA patterns. Each increment must justify its latency and parameter cost.

For robustness analysis, we assess performance across DGA subclasses, sensitivity to unseen or rare families, and degradation under simulated distribution shifts. The primary shift simulation is a leave-one-family-out evaluation protocol: models are trained on all but one malware family and tested on the held-out family, repeated across all families. This directly measures generalization to novel DGA behavior, which is the central concern of the project.

Initial Benchmarking and Ablation Studies

The problem is split into two sub-problems.

Sub-problem A: Binary classification (DGA vs. benign)

Sub-problem B: Multi-class classification (malware family, e.g. cryptolocker, nivdort)

Loss Functions

For sub-problem A, the goal is to show that low FPR and low latency is feasible. We use binary cross-entropy (BCE) as the primary loss function. BCE measures the dissimilarity between the model's predicted probability and the true binary label for each sample, penalizing confident wrong predictions heavily via a logarithmic penalty. Minimizing BCE corresponds to maximum likelihood estimation under a Bernoulli probabilistic model. For linear classifiers it is convex, guaranteeing a unique global minimum. While deep models are non-convex, BCE still provides a smooth and well-behaved optimization landscape suitable for gradient-based training. We acknowledge that minimizing BCE alone does not explicitly optimize FPR, so we will additionally experiment with weighted binary cross-entropy, which assigns a higher penalty to false positive errors by scaling the loss contribution of benign samples, and with focal loss (Lin et al., 2017), which introduces a modulating factor to the standard BCE that down-weights easy, confidently-classified examples and focuses training on hard ambiguous ones. Post-training threshold calibration — adjusting the decision boundary away from the default 0.5 probability cutoff — will be applied as an additional mechanism to directly control the FPR-recall trade-off at deployment time.

For sub-problem B, the goal is to provide a second point of confirmation that the features identified in sub-problem A form a clear class separation structure, validating sub-problem A. The primary loss is categorical cross-entropy (CCE), which generalizes BCE to K mutually exclusive classes by measuring the dissimilarity between the predicted softmax probability distribution and the one-hot encoded true class label. CCE is the standard loss for multi-class softmax-output classifiers and is chosen here for its probabilistic interpretability and compatibility with all neural architectures in our model suite. For the SVM-based approach we use multi-class hinge loss, which penalizes any prediction where the margin — the gap between the score of the correct class and the highest-scoring incorrect class — falls below a fixed threshold of 1. Unlike CCE, which optimizes class probabilities, hinge loss directly optimizes decision boundary geometry by maximizing this margin, pushing boundaries as far as possible from training data. Comparing CCE and hinge loss therefore reveals whether family separation is a probabilistic or geometric phenomenon in the learned feature space. If the binary classifier in sub-problem A learns genuinely robust lexical features, those features should produce well-separated family clusters here. Weak family separation would suggest the binary classifier is exploiting shallow surface patterns rather than stable structural features.

Models

For sub-problem A, we train and compare the following model families in order of increasing complexity. Each model must demonstrate a measurable improvement in generalization over the previous to justify its additional parameter and latency cost.

1. Logistic Regression with N-gram Features serves as the baseline. Character-level n-gram TF-IDF (Term Frequency-Inverse Document Frequency) vectors are constructed by treating each domain string as a short document and computing weighted counts of all contiguous character subsequences of length $n = 2$ to 4. TF-IDF downweights n-grams that appear frequently across all domains (and are therefore uninformative) while upweighting those that are distinctive to specific samples. These vectors are fed into a logistic regression classifier, establishing a lightweight reference point with interpretable feature weights and negligible inference latency.
2. Gradient Boosted Trees (XGBoost) uses hand-crafted lexical features — domain length, Shannon entropy (a measure of character randomness), vowel ratio, digit ratio, n-gram frequencies, and longest consonant run — as inputs. Shannon entropy is particularly diagnostic of DGA behavior, as algorithmically generated domains tend to have higher character-level entropy than human-readable ones. XGBoost builds an ensemble of decision trees sequentially, each correcting the residual errors of the previous, producing a strong classifier from many weak ones. This tests whether classical feature engineering can close the gap with neural methods while remaining highly deployment-efficient.
3. Character-level Transformer uses a lightweight encoder with multi-head self-attention, which allows every character position to directly attend to every other position in a single operation, capturing long-range dependencies without the sequential bottleneck of RNNs. We can use the pre-trained CANINE transformer from google. This tests whether self-attention over character positions captures interactions that sequential or convolutional models miss.
4. Autoencoder (Anomaly Detection) consists of an encoder that compresses a domain string into a low-dimensional latent representation, and a decoder that attempts to reconstruct the original character sequence from that representation. The model is trained exclusively on benign domain strings, so it learns a compact model of legitimate domain structure. At inference time, reconstruction error — the dissimilarity between the input and its reconstruction — serves as an anomaly score: benign domains reconstruct well and score low, while DGA domains, being out-of-distribution, reconstruct poorly and score high. This reframes sub-problem A as unsupervised anomaly detection, motivated by the core robustness concern that a model trained only on benign data may generalize better to unseen DGA families than a supervised model trained on a fixed set of known malware. Training loss is BCE over character tokens; at inference, reconstruction error is thresholded to produce a binary decision.

For sub-problem B, we use a Multi-class Linear SVM (Support Vector Machine) with one-vs-rest or Crammer-Singer multi-class hinge loss, trained on the same n-gram or hand-crafted features as the baseline. An SVM seeks the maximum-margin hyperplane separating classes in feature

space — the decision boundary that is as far as possible from the nearest training examples of each class. A kernel SVM using the RBF (Radial Basis Function) kernel, which implicitly maps inputs into a higher-dimensional space to enable non-linear separation, will be explored if linear separation proves insufficient.

Possible Methods of Analysis

For sub-problem A:

- AUC-ROC (Area Under the Receiver Operating Characteristic) curve, which plots the True Positive Rate against the False Positive Rate across all possible classification thresholds, giving a threshold-independent measure of a classifier's ability to separate classes - a random classifier scores 0.5 and a perfect classifier scores 1.0. This is chosen because it captures the full FPR-recall trade-off rather than performance at a single operating point. We also produce a latency vs. performance trade-off plot and a threshold vs. FPR curve to guide deployment decisions.

For sub-problem B:

- Margin distribution plot, which shows the signed distance of each sample from the SVM decision boundary — points with large positive margin values are confidently and robustly classified, while points near zero are ambiguous.
- Centroid distance matrix, which computes pairwise distances between per-family embedding centroids to reveal which malware families are geometrically proximate and may share lexical generation strategies.
- PCA (Principal Component Analysis) plot, which projects high-dimensional feature vectors onto their two highest-variance directions for a fast, linear overview of class separability before applying more expensive non-linear projections.

For embedding visualization across neural models (CNN, RNN, Transformer), we extract the penultimate-layer embedding — the fixed-length vector produced immediately before the classification head — for all test samples and apply t-SNE (t-distributed Stochastic Neighbor Embedding) and UMAP (Uniform Manifold Approximation and Projection) projections to two dimensions.

Both methods preserve local neighborhood structure, with t-SNE emphasizing tight cluster formation and UMAP better preserving global inter-cluster distances; using both provides a more complete picture than either alone. Points are colored by binary label and by malware family label to assess whether family structure emerges in a model trained only on the binary task.

Attention heatmaps (Transformer only) visualize per-character attention weights, providing a direct interpretable view of which substrings are most diagnostic of DGA behavior. For the autoencoder, reconstruction error distributions are plotted separately for benign domains, known DGA families, and held-out DGA families to assess the viability of the anomaly detection framing and to identify the threshold sensitivity at the overlap region.

Evaluation Summary

Primary metrics:

- False Positive Rate (FPR, the fraction of benign domains incorrectly flagged as DGA — the operational priority, as excessive false alerts render a detection system unusable in practice)
- Precision (the fraction of flagged domains that are truly DGA)
- Recall (the fraction of all DGA domains that are successfully detected)
- F1-score (the harmonic mean of Precision and Recall, balancing the two without allowing either to dominate)
- AUC-ROC.

Operational metrics:

- Average and tail inference latency (the latter capturing worst-case response time under load)
- Parameter count

Robustness metrics:

- Per-family F1 variance (measuring consistency of detection across malware families rather than aggregate performance)
- Leave-one-family-out generalization gap (the drop in F1 between seen and held-out families)
- Directly quantifying how well the model handles novel DGA behavior)
- Reconstruction error separation for the autoencoder.

Visualization outputs:

- t-SNE and UMAP embedding plots
- centroid distance matrix
- attention heatmap
- margin distribution plots
- reconstruction error distributions.

References

Charles Givre (2021) DGA Dataset <https://www.kaggle.com/datasets/gtkcyber/dga-dataset>

Manasrah A. M., Khmour T.,Freehat R. (2022) DGA-based botnets detection using DNS traffic mining, Journal of King Saud University - Computer and Information Sciences.
<https://doi.org/10.1016/j.jksuci.2022.03.001>

Jeremiah D., Rafiq H.,Ta V. T., Usman M., Raza M., Awais M. (2025) NIOM-DGA: Nature-inspired optimised ML-based model for DGA detection, Computers & Security.
<https://doi.org/10.1016/j.cose.2025.104561>